WILLIAM DUAN

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EDUCATION

Ph.D. in Economics, University of British Columbia 2020 – 2027 (expected)
Co-supervisors: Hao Li and Vitor Farinha Luz. Committee member: Wei Li.

M.A. in Economics, University of British Columbia 2019 – 2020

B.A., First Class Honours in Economics, McGill University 2015 – 2019
Minor in Mathematics and Computer Science. GPA 3.92.

FIELDS

Primary: Microeconomic Theory
Secondary: Mechanism Design; Information Economics

JOB MARKET PAPER

Selling Mechanisms under Misspecified Learning
Presented at the Canadian Economics Association Annual Meetings, Vancouver, May 2026.

Buyers in a repeated selling mechanism do not observe its allocation rule. Each holds a possibly misspecified parametric model of his win probability, bids optimally under his current belief about the model's parameter, and learns by Bayesian updating under that model from allocation outcomes alone. The mechanism generates those outcomes, so it shapes what buyers come to believe. For a given model and belief, we ask when there exists a mechanism that sustains that belief while buyers keep learning, and how such a mechanism does it. Such a mechanism exists when the win probability predicted by the model responds strongly enough to a change in belief, relative to how bidding behaviour responds. We give sufficient conditions of this kind for convergence from every initial belief with a single buyer, and for convergence from nearby beliefs with two buyers, where feasibility ties the win probability each faces to the other's bids. To extend the basin of attraction beyond nearby beliefs, we work with discrete valuations and give a computational criterion in the form of a linear program, whose strictly feasible solutions yield mechanisms that sustain the belief from every more optimistic belief.

WORKING PAPERS

Ex-ante Heterogeneity in Directed Search Games
Shows that the standard symmetric equilibrium in directed search with finitely many workers is fragile to small cross-worker differences in skill distributions, and characterizes a robust equilibrium that always exists and connects it to the tractable symmetric benchmark.

FELLOWSHIPS, SCHOLARSHIPS AND AWARDS

Agustín Beltrán Award for Excellence in Teaching, Vancouver School of Economics 2024, 2025, 2026
SSHRC Doctoral Fellowship 2020 – 2024
President's Academic Excellence Initiative PhD Award, UBC 2020 – 2024
A. D. Scott Fellowship in Economics, UBC 2020
Faculty of Arts Graduate Award, UBC 2019 – 2020
Hugh Brock Scholarship, McGill University 2015 – 2018
J. W. Birch Economics Undergraduate Award, McGill University 2017
Dean's Honour List, McGill University 2015 – 2017

TEACHING EXPERIENCE

Teaching Assistant, Vancouver School of Economics, UBC 2019 – 2026

- ECON 600/601 Advanced Microeconomic Theory I/II (Ph.D. core) 2021 – 2026
  Agustín Beltrán Award for Excellence in Teaching, 2024, 2025, 2026.
  Co-designed ECON 601 problem sets, 2025, 2026.
- ECON 101 Principles of Microeconomics 2020, Summer 2025
- ECON 514 Information and Incentives 2021
- ECON 303 Intermediate Microeconomic Analysis II 2020
- ECON 355 Introduction to International Trade 2019

Instructor, VSE Math Camp (M.A. and Ph.D. sessions) Summer 2026

Designed the Ph.D. sessions with Prof. Jesse Perla and wrote their slides and practice problems. For the M.A. sessions, which ran on recorded faculty lectures, led the problem sessions and questions.

RESEARCH EXPERIENCE

Research Assistant to Prof. Vitor Farinha Luz, University of British Columbia 2022

Research Assistant to Prof. Hassan Benchekroun, McGill University 2017 – 2019

Theoretical derivations for dynamic programming models and numerical simulation of models in international trade and resource economics.

SKILLS

Languages: English and Mandarin Chinese (fluent); French (basic)
Software: Python, R, Julia, MATLAB, Mathematica, Java; LaTeX, Quarto, Git

REFERENCES

Prof. Hao Li (co-supervisor)
Vancouver School of Economics, UBC
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Prof. Vitor Farinha Luz (co-supervisor)
Vancouver School of Economics, UBC
vitor.farinhaluz@ubc.ca

Prof. Wei Li
Vancouver School of Economics, UBC
Wei.Li@ubc.ca

Prof. Jesse Perla
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Last updated: October 2026